

Inferring Latent Bitcoin Market Regimes with Hidden Markov Models

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Abstract—This project investigates whether hidden Markov models can infer meaningful latent Bitcoin market regimes from weekly market observations. We compare price-only, macro, trader-style technical analysis, and psychology-enhanced feature families. The strongest model combines price information with psychology-linked variables, including the Crypto Fear and Greed Index, momentum, and drawdown. Results suggest that psychology-linked features improve regime inference more than macro-only and trader-only feature sets in this weekly setting. The inferred regimes are economically interpretable and align with consolidation (sideways movement of the market), greed or bull (uptrend), and fear or stress (downtrend) market conditions. Additional evaluation against simple baselines and a masked-feature recovery test supports the usefulness of temporal latent-state modeling for short-term Bitcoin regime analysis.

Index Terms—Hidden Markov Model, Bitcoin, market regimes, sentiment, behavioral finance, time series

I. INTRODUCTION

Bitcoin (BTC) is well known for being unpredictable and sporadic, BTC’s price is often volatile due to abrupt shifts in trend and investor sentiment. In practice, traders and analysts often describe these periods informally as bull phases, panic sell-offs, recovery periods, or choppy consolidation zones. However, many standard forecasting approaches attempt to predict the next return directly without explicitly modeling such latent market states. This can produce short-term predictions, but it often provides limited insight into the underlying condition of the market.

This paper approaches the problem differently by modeling BTC as a sequence of hidden regimes. Instead of assuming that all weeks are generated by the same process, the model assumes that the market transitions between latent states, where each state produces different observable patterns in return, volatility, trading activity, and psychology-linked indicators. This makes a hidden Markov model a suitable probabilistic graphical model for the task, since the hidden state can encode the unobserved regime while the observed variables provide noisy evidence about it.

The central question I would like to answer is whether psychology-linked variables improve BTC regime inference relative to price-only, macro, and trader-style technical indicators. In this project, psychology-linked variables refer to features such as the Crypto Fear and Greed Index, momentum, and drawdown, which are intended to reflect market mood and overall stress. This framing is useful because BTC markets

are often described in psychological terms, in my experience BTC’s price has always had more emotional reactions compared to the rest of the stock market which is relatively more stable and such interpretation is difficult to capture with purely static or purely price-based methods.

A second objective is to evaluate whether the inferred regimes are useful beyond description. To test this, the paper studies two downstream tasks. First, it measures whether the inferred regimes improve short-term directional prediction relative to simple baselines. Second, it performs masked-feature recovery, or cover-up testing, in which known variables are hidden and reconstructed from the remaining observations through the learned latent-state structure. This allows the model to be evaluated both as a predictive model and as a generative probabilistic model.

The contributions of this work are fourfold. First, it presents a weekly hidden Markov model framework for latent BTC regime inference. Second, it compares five feature families spanning price-only, psychology-linked, macro-financial, trader-style, and combined inputs. Third, it evaluates the model against simpler baselines including logistic regression and Gaussian mixture clustering. Fourth, it introduces a masked-feature recovery experiment to test whether the model captures meaningful joint structure among psychology-linked variables.

A. Why Regime Inference Matters

A major difficulty in BTC analysis is that market behavior is not stable over time. A strategy or predictive rule that performs reasonably during a bullish phase may fail during a stressed or high-volatility phase because the underlying data-generating process has changed. This creates a practical problem for both forecasting and interpretation. A single global model can smooth over these shifts and hide the fact that very different market conditions are being treated as if they were generated by one stationary process.

The hidden Markov model addresses this problem by explicitly representing the market as a sequence of latent regimes. Instead of asking only whether the next week will be positive or negative, the model first asks what kind of market condition is currently active. This separates two tasks that are often mixed together in simpler approaches: identifying the current regime and making a forecast conditional on that regime. In this sense, the model provides both a probabilistic description

of market structure and a practical basis for conditional prediction.

II. DATA AND FEATURE FAMILIES

The dataset consists of weekly observations constructed from daily market and macro-financial series beginning in February 2018. Weekly frequency was selected mainly to align BTC observations with slower-moving sentiment and macro-financial variables, and to keep the inferred regimes interpretable under a limited sample size. This choice also reduces the mismatch between continuously traded BTC data and lower-frequency auxiliary indicators.

The core BTC market variables are weekly log return, 4-week rolling volatility, and log volume change. These variables form the price-only baseline and provide direct information about directional movement, market turbulence, and participation intensity. On top of these, additional feature families were designed to test distinct sources of information.

The psychology feature family extends the price variables with the Crypto Fear and Greed Index, 4-week momentum, and 12-week drawdown. Fear and Greed provides a direct sentiment proxy, momentum reflects persistence and trend-following behavior, and drawdown captures stress relative to a recent peak. Together, these variables allow the inferred hidden states to be interpreted not only statistically but also behaviorally.

The macro feature family augments the price variables with dollar-index return and the weekly change in the 10-year Treasury yield. These variables were included to test whether broader financial conditions improve BTC regime inference. The trader-style feature family adds moving-average spread and price relative to a longer moving average, resembling indicators used in common technical analysis workflows. Finally, a compact combined family mixes a small set of variables drawn from multiple groups to test whether combining information sources improves performance.

A. Data Quality and Ethical Considerations

Several data quality limitations warrant explicit discussion. The Crypto Fear and Greed Index is a proprietary composite of multiple sub-indicators, including social media sentiment, market momentum, and surveys. Its weighting methodology is not fully transparent, and the index is known to have changed construction over time, introducing potential non-stationarity in what the variable measures. The index was forward-filled on weeks with missing daily observations, which introduces a stale-value assumption: weeks with no new sentiment reading inherit the most recent past value rather than a contemporaneous one.

The available weekly sample is limited in size due to the start of the Fear and Greed series in 2018. This constrains the number of complete market cycles observed and increases the risk that the learned regimes reflect the specific macro conditions of the 2018–2024 period rather than stable structural properties of BTC markets. Results should therefore

be interpreted as exploratory and period-specific rather than general.

Since the project uses public aggregated financial and sentiment data, no personal data is involved. However, a probabilistic regime model can easily be misrepresented as deterministic trading advice, especially in retail contexts where uncertainty is often under-communicated. Results in this project are presented as exploratory modeling outcomes and not as financial advice. Another consideration that I have personally seen asks the question of should we even be using 'bots' to trade. Many a time bots flood the market during quiet times and influence price heavily due to the sheer number of them trading, this in turn affects price and can influence other 'real' peoples trades. We are adding to this issue of sculpting the market if we misuse and abuse trading bots. However since we do use the weekly time frame this issue is minimized. It is important to keep in mind we are using computers to influence real life decisions and real money if we choose to use this model on a live chart.

III. METHODOLOGY

A. Probabilistic Graphical Model Formulation

Let z_t denote the hidden market regime at week t and let $\mathbf{x}_t$ denote the observed feature vector. The hidden Markov model is a dynamic Bayesian network in which the current hidden regime depends only on the previous regime (the Markov property) and the observed variables are conditionally independent given the current regime:

$$P(z_{1:T}, \mathbf{x}_{1:T}) = P(z_1) \prod_{t=2}^T P(z_t | z_{t-1}) \prod_{t=1}^T P(\mathbf{x}_t | z_t). \quad (1)$$

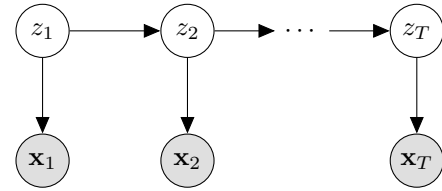


Fig. 1. Plate-free graphical model of the HMM. Shaded nodes ($\mathbf{x}_t$) are observed; unshaded nodes (z_t) are latent market regimes. Edges encode the Markov transition structure and conditional emission independence.

The graphical structure is shown in Fig. 1. The key conditional independence assertion encoded by the graph is that each observation $\mathbf{x}_t$ depends only on the current hidden state z_t , and is conditionally independent of all other observations and hidden states given z_t . This is what distinguishes the HMM from a Kalman filter, which assumes linear-Gaussian dynamics in a continuous latent space, and from a static Gaussian mixture model, which drops the transition structure entirely. The HMM occupies a useful middle ground: it captures temporal dependence through the Markov chain while keeping inference tractable.

B. Parameter Learning and Inference

Model parameters—the initial state distribution π , the transition matrix A where $A_{ij} = P(z_t = j \mid z_{t-1} = i)$, and the Gaussian emission parameters (μ_k, Σ_k) for each state k —are estimated by maximizing the marginal log-likelihood $\log P(\mathbf{x}_{1:T})$ using the Baum-Welch algorithm, which is an instance of expectation-maximization. The E-step computes the forward-backward messages:

$$\alpha_t(k) = P(\mathbf{x}_{1:t}, z_t = k), \quad (2)$$

$$\beta_t(k) = P(\mathbf{x}_{t+1:T} \mid z_t = k), \quad (3)$$

from which the posterior state occupancies and pairwise state transitions are computed. The M-step then updates parameters in closed form. This procedure is guaranteed to converge to a local optimum of the likelihood. Since the objective is non-convex, multiple random restarts were used and the solution with the highest training log-likelihood was selected.

Once parameters are fixed, the most probable latent state sequence is decoded using the Viterbi algorithm, which solves $\arg \max_{z_{1:T}} P(z_{1:T} \mid \mathbf{x}_{1:T})$ via dynamic programming. This is distinct from the soft posterior $P(z_t \mid \mathbf{x}_{1:T})$ computed by the forward-backward pass; Viterbi provides a single globally consistent state sequence rather than marginal probabilities.

C. Model Variants and Design Choices

Gaussian HMMs with diagonal covariance were fitted. Full covariance was considered but ruled out due to the limited weekly sample size relative to the number of free parameters. A left-to-right (Bakis) transition structure was also considered, but rejected because BTC regimes are not ordered in a fixed progression; the market can return to a previous regime. Both three-state and four-state models were tested for each feature family. Chronological train-test splitting was used throughout so that future weeks were never observed during training. Feature scaling was fitted on the training period only and applied unchanged to the test period.

IV. EVALUATION PROTOCOL

Evaluation was intentionally designed to be broader than a single accuracy score, covering predictive usefulness, interpretability, temporal realism, and generative quality.

A. Likelihood-Based Model Fit

Train and test log-likelihood, AIC, and BIC were recorded for each model. These are interpreted cautiously across feature families, since models with different observation dimensionality are not directly comparable through raw likelihood values. Likelihood-based measures were therefore used primarily to compare state counts within the same feature family.

B. Feature-Family Comparison

The first evaluation compares five feature families under both three-state and four-state configurations using out-of-sample accuracy and balanced accuracy for next-week direction.

TABLE I
BEST-PERFORMING CONFIGURATION WITHIN EACH FEATURE FAMILY.

Feature family	Features	States	Accuracy	Balanced accuracy
Price only	3	4	0.548	0.530
Price + psychology	6	3	0.583	0.572
Price + macro	5	3	0.464	0.467
Price + trader	5	4	0.500	0.505
Compact combined	7	3	0.548	0.500

TABLE II
COMPARISON OF THE BEST HMM AGAINST NON-HMM BASELINES ON THE TEST SET.

Model	Accuracy	Balanced accuracy
Majority class	0.548	0.500
Last sign	0.524	0.519
Logistic regression	0.524	0.535
GMM (3 clusters)	0.548	0.543
HMM psychology (3 states)	0.583	0.572

C. Baseline Comparison

The second evaluation compares the best-performing HMM against a majority-class rule, a last-sign rule, logistic regression, and a Gaussian mixture model. The GMM baseline uses identical features but no temporal dependence, providing a direct test of whether the Markov structure adds value.

D. Cover-Up Testing

A masked-feature recovery experiment hides a known psychology-linked feature on the test set and reconstructs it using state-conditioned means estimated from training data. Reconstruction quality is measured using MAE and RMSE against a global-mean baseline. This evaluates the HMM as a generative model rather than only a directional predictor.

E. Sensitivity Analysis

A useful regime model should not be judged by a single best score. Both three-state and four-state models were tested per feature family to assess interpretability versus granularity. A model with too few states may merge meaningfully different market conditions, while too many states risk fragmenting the data into clusters that are hard to interpret. Across multiple random restarts, the best-fitting psychology-enhanced models tended to recover qualitatively similar state profiles, with label permutations but stable behavioural interpretations. The comparison across five feature families also acts as a broader sensitivity test: if conclusions changed entirely under small feature variations the inferred regimes would be untrustworthy. Instead, the experiments showed a stable pattern: price-only remained competitive, trader-style was informative but weaker, macro performed poorly, and psychology-enhanced was consistently the strongest - strengthening the claim that the detected regimes are not artifacts of one arbitrary feature choice.

TABLE III
MASKED-FEATURE RECOVERY ON THE TEST SET.

Masked feature	HMM MAE	Mean MAE	HMM RMSE	Mean RMSE
Fear & Greed	15.440	19.964	18.467	22.543
4-week momentum	0.092	0.094	0.109	0.129
12-week drawdown	0.077	0.102	0.093	0.111

TABLE IV
INTERPRETATION OF THE THREE LATENT STATES IN THE
BEST-PERFORMING PSYCHOLOGY-ENHANCED HIDDEN MARKOV MODEL.

State	Mean return	Fear/Greed	Momentum	Drawdown	Interpretation
State 0	-0.008	53.96	0.039	-0.078	Neutral or recov
State 1	0.081	68.56	0.213	-0.000	Greed or bull
State 2	-0.025	25.01	-0.109	-0.272	Fear or stress

V. RESULTS AND ANALYSIS

Within feature families, likelihood-based metrics were consistent with the final state-count choices reported here, but were not used as the main basis for cross-family ranking because the observation dimensionality differs across models. The strongest feature family was the psychology-enhanced model with three latent states, achieving an out-of-sample accuracy of 0.583 and balanced accuracy of 0.572 (Table I). The macro-only family performed worst at 0.464 accuracy, suggesting that dollar-index return and Treasury yield changes are too broad relative to BTC-specific dynamics at weekly resolution. The trader-style family reached 0.500 accuracy, informative but weaker than psychology-linked variables, indicating that moving-average relationships capture only part of the latent regime structure. The compact combined family (0.548 accuracy, seven features) offered no improvement over the psychology-only configuration despite the larger feature set, consistent with the view that adding variables under limited sample size can hurt rather than help. The price-only baseline also reached 0.548, underscoring that feature quality matters more than quantity in this setting.

The decoded states are economically interpretable. Table IV summarizes the three learned regimes. State 1 corresponds to a greed or bull regime with high positive return, elevated Fear and Greed, strong momentum, and negligible drawdown. State 2 corresponds to a fear or stress regime with negative return, low Fear and Greed, negative momentum, and deep drawdown. State 0 lies between these extremes and is best interpreted as a neutral or recovery regime.

Table V shows the transition structure. The fear or stress state is the most persistent (self-transition probability 0.899), consistent with the empirical observation that stressed markets tend to remain stressed for extended periods. The greed or bull state does not transition directly into the fear state, supporting a more realistic bull-to-neutral-to-fear progression. This temporal coherence is an important property that distinguishes the HMM from the GMM baseline: the GMM assigns the same cluster to each week independently, and so cannot enforce this kind of regime pathway.

TABLE V
TRANSITION MATRIX FOR THE BEST-PERFORMING THREE-STATE
PSYCHOLOGY-ENHANCED HIDDEN MARKOV MODEL. ENTRIES OF 0.000
ARE ROUNDED TO THREE DECIMAL PLACES AND MAY NOT BE EXACTLY
ZERO.

	To State 0	To State 1	To State 2
From State 0	0.687	0.223	0.090
From State 1	0.418	0.582	<0.001
From State 2	0.101	<0.001	0.899

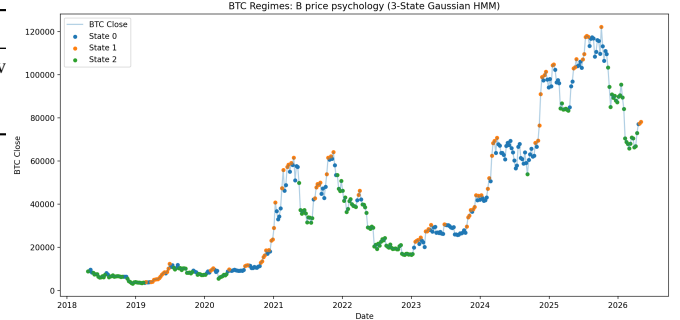


Fig. 2. BTC closing prices with decoded states from the best-performing three-state psychology-enhanced hidden Markov model.

What makes these transition dynamics particularly interesting is that they mirror how market participants actually describe regime changes. Traders rarely say the market “instantly crashed from euphoria to panic.” Instead, they describe a slow deterioration: prices stop making new highs, momentum fades, and fear gradually overtakes greed before a proper sell-off begins. The transition matrix captures exactly this progression probabilistically. The greed state softens into neutral with probability 0.418, and direct transition from greed to fear is effectively zero after rounding, so fear is reached mainly through the neutral state. This is not a constraint imposed on the model — it is something the model learned from the data.

The baseline comparison confirms that the HMM (0.583 accuracy, 0.572 balanced accuracy) outperforms all simpler alternatives (Table II). The majority-class rule achieves 0.548, reflecting the slight upward bias in the test period. Logistic regression reaches only 0.524, indicating that the features alone, fed to a direct discriminative model, are insufficient. The GMM with the same six features achieves 0.548 but with a balanced accuracy of 0.543 — higher balanced accuracy than the majority baseline but still below the HMM. This gap specifically isolates the value of the Markov transition structure, since the feature set is identical between the GMM and the HMM.

The cover-up results show that state-conditioned reconstruction outperforms the global-mean baseline for all three masked features (Table III). The improvement is largest for Fear and Greed (HMM MAE 15.44 versus baseline 19.96, a 23% reduction) and 12-week drawdown (HMM MAE 0.077 versus 0.102, a 25% reduction). These two features align most strongly with the greed and fear state definitions, so

it is unsurprising that knowing the latent state substantially improves their reconstruction. The gain for 4-week momentum is smaller but still positive (HMM MAE 0.092 versus 0.094), suggesting that momentum carries some information shared with the other features once the latent state is conditioned upon.

VI. DISCUSSION

A. Comparison with Trader-Style Reasoning

An important part of this project was comparing the HMM against the kinds of indicators that traders commonly use in practice. Moving-average relationships, in particular, are widely used to identify trend continuation and reversal zones. These tools are appealing because they are simple, visual, and easy to interpret. However, they are deterministic heuristics - a moving-average crossover gives a signal but does not tell you the probability that the market is in a fear-driven or greed-driven regime, or how long that regime is likely to persist.

The trader-style family achieved 0.500 accuracy, weaker than the psychology-enhanced model. This is an interesting finding because it suggests that BTC regime structure is not fully captured by trend indicators alone. Moving averages describe where price is relative to its own recent history, but they do not directly encode whether participants are psychologically complacent, fearful, or euphoric. A market can be above its 12-week moving average and still be in a fear regime if sentiment is collapsing and drawdown is deepening — which is exactly the kind of distinction the HMM can detect when psychology-linked variables are included.

This is also where the HMM offers something a trader’s checklist cannot: a probabilistic description of market regime rather than a purely deterministic signal.

B. Why Market Psychology Drives Regime Structure

The stronger performance of psychology-linked variables is not surprising when you consider how BTC markets actually work. Unlike traditional equity markets, BTC has no earnings, no dividend yield, and no fundamental anchor in the classic sense. BTC price is heavily influenced by collective belief and crowd behavior. This means that Fear and Greed, momentum, and drawdown appear to be among the most informative variables for regime transitions in this setting.

The cover-up experiment provides indirect evidence for this. The Fear and Greed Index and 12-week drawdown were the easiest features to reconstruct from latent states (23% and 25% MAE reductions respectively). This means the learned states are genuinely organized around psychological structure: once you know the state, you already know a great deal about how fearful or greedy the market is. Momentum was harder to reconstruct, which makes sense — momentum is a transitional feature that changes within a regime, not a stable characteristic of one.

The macro-only family performed worst at 0.464, below the majority-class baseline. This reinforces the view that BTC is not well-described by traditional macro variables at weekly frequency. Dollar strength and Treasury yields matter

for risk-on or risk-off behavior in institutional portfolios, but BTC’s retail-dominated weekly dynamics are driven more by sentiment cascades than by macro fundamentals.

C. Limitations

The Gaussian emission assumption imposes symmetric, light-tailed observations on each state. BTC returns are heavy-tailed, so extreme episodes — the March 2020 crash, the 2021 peak, the 2022 collapse — may be represented imperfectly. A Student-t HMM would be a natural extension. The limited sample also constrains the number of complete market cycles observed, increasing period-specificity. The masked-feature reconstruction design was intentionally simple: state-conditioned mean imputation rather than full conditional inference. It serves as a test of joint structure, not an optimal recovery procedure. Future work could explore richer sentiment sources, BTC dominance as an additional feature, or longer-horizon regime tasks. Running these same experiments with other tickers like Ethereum could also be interesting since smaller market cap cryptocurrencies tend to follow BTC’s price structure but in a somewhat delayed manner due to money moving from large market cap cryptocurrencies into smaller market cap cryptocurrencies, which is also reflected through the BTC dominance indicator.

VII. CONCLUSION

This paper presented a hidden Markov model approach for inferring latent BTC market regimes from weekly data. Framed as a dynamic Bayesian network, the HMM captures temporal dependence through a Markov chain over hidden states, with Gaussian emissions observed at each step. Parameters were estimated via the Baum-Welch algorithm and regimes were decoded with the Viterbi algorithm. Among the compared feature families, a psychology-enhanced HMM achieved the strongest out-of-sample performance (0.583 accuracy, 0.572 balanced accuracy) and yielded clear market-psychology interpretations. The best model separated the market into neutral or recovery, greed or bull, and fear or stress regimes supported by both feature summaries and transition structure.

The HMM outperformed all simpler baselines, including logistic regression (0.524) and Gaussian mixture clustering (0.548), and recovered masked psychology-linked features with up to 25% lower MAE than a global-mean baseline. The improvement over the GMM confirms that temporal latent-state structure contributes beyond static clustering. From a practical perspective, the project suggests that understanding BTC in terms of latent behavioral regimes may be more informative than relying on price-only signals or common trader heuristics in isolation.